

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks: _____

Computer Networking And Data Communication(LAB)

Project# 02

Title: Computer Network Scenario

Submitted To: Mr Ahsan Abbas

Student's Names	Registration No.
Syeda Nalain Zaineb Bukhari	24108233
Tehniat Khan	24108235
Aiman Nasir	24108202
Malika Naz	24108212
Safia Alam	24108229

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

1. Introduction

In the 21st century, colleges and universities rely heavily on advanced information and communication technologies to support their daily operations. Network infrastructure has become the backbone of academic institutions, enabling data sharing, collaboration, online learning, and access to digital resources. An efficient and secure Local Area Network (LAN) is no longer a luxury—it is an essential requirement for modern educational environments.

The College Network Scenario (CNS) project focuses on designing a reliable, high-performance, and secure LAN architecture for a college campus. This project aims to create a model that integrates different departments into a unified communication system, enhancing coordination, information sharing, and data management across the entire institution.

In a typical college setting, departments such as Information Technology, Computer Science, Administration, Examination Control, and Finance rely on digital tools to handle academic records, staff data, research outputs, and online assessments. A well-designed LAN ensures that all these entities can seamlessly communicate and exchange data without interference or bottlenecks.

The CNS project proposes the development of a network topology that supports both wired and wireless connectivity, ensuring accessibility for staff and students. The new design uses Cisco-based technologies, known for their reliability, scalability, and advanced management capabilities. This network also incorporates Virtual Local Area Networks (VLANs) for segmentation, Routing Information Protocol (RIP) for dynamic routing, and Cisco Unified Computing System (UCS) for centralized control and server integration.

By implementing this model, the project seeks to modernize the college's existing infrastructure, reduce operational costs, enhance network security, and prepare the institution for future expansions such as cloud integration, e-learning platforms, and smart campus applications.

1.1 Background

The existing network in the college operates on an outdated infrastructure that limits its performance, scalability, and reliability. The current system is primarily based on static IP addressing, with most devices manually configured. This manual setup leads to administrative difficulties, especially when devices are added, replaced, or relocated.

The network devices—primarily older unmanaged switches—lack the ability to monitor or control network traffic efficiently. Because there is no VLAN configuration, all devices share a single broadcast domain. This increases the risk of data collisions, slows down performance, and compromises security, as sensitive administrative data and general student traffic coexist on the same network segment.

Additionally, the absence of remote access mechanisms means that network administrators cannot monitor or troubleshoot the network from off-site locations. The lack of a firewall also makes the system vulnerable to external attacks and internal misuse.

Another major challenge is resource fragmentation. The college maintains separate servers for each department—such as Examination, IT, and Library—resulting in redundant power usage, space constraints, and high maintenance costs.

Recognizing these shortcomings, the CNS project introduces a modernized infrastructure that incorporates Layer 2 and Layer 3 Cisco switches, dynamic routing protocols, and centralized server management. By migrating to this advanced architecture, the college can achieve:

- Efficient traffic segmentation via VLANs,
- Dynamic routing and automated network updates using RIP,
- Remote and centralized network management through Cisco UCS, and
- Improved energy efficiency by consolidating server operations.

This transition represents a major leap forward, aligning the college's IT infrastructure with international standards for academic institutions.

1.2 Scope of the Project

The scope of the College Network Scenario project extends across the entire network life cycle—from conceptualization to simulation and evaluation. The project focuses on building a model campus LAN that could serve as a template for real-world deployment.

Core Components of the Project Scope

1. **Planning and Analysis**
 - Assessing existing infrastructure and identifying limitations in performance, security, and scalability.
 - Determining network requirements such as number of users, devices, and inter-departmental connectivity.
2. **Network Design**
 - Developing a detailed topology using Cisco Packet Tracer that includes routers, switches, servers, and wireless access points.
 - Implementing VLANs for traffic segmentation and RIP for inter-department communication.
3. **Implementation (Simulation Stage)**
 - Configuring all devices virtually in Cisco Packet Tracer.
 - Assigning IP addresses, subnetting, and setting up routing tables.
 - Testing data transmission across departments and validating connectivity.
4. **Security Configuration**
 - Integrating firewall concepts, access control lists (ACLs), and VLAN isolation.
 - Introducing UCS-based centralized authentication for server access.
5. **Testing and Evaluation**
 - Measuring network performance using throughput and latency metrics.
 - Testing scenarios such as server access, web hosting, and FTP file transfers.
6. **Documentation and Proposal Development**
 - Preparing a Request for Proposal (RFP) including hardware specifications, cost estimation, and deployment plan.

Although this project is conducted through simulation, the design is realistic and adaptable for physical implementation with Cisco devices. The model can easily scale to accommodate new departments, hostels, or research centers.

1.3 Objectives

The CNS project sets out several objectives that guide its overall design and implementation:

1. **Modernization of Infrastructure**
Upgrade the existing outdated network to a high-performance system capable of supporting modern technologies like VoIP, cloud storage, and Wi-Fi 6.
2. **Enhanced Security and Control**
Implement VLANs, firewalls, and role-based access through UCS to protect sensitive data from unauthorized access.
3. **Centralized Management**
Utilize Cisco UCS to provide a single management interface for servers, switches, and routers, simplifying administrative tasks.
4. **Reliability and Scalability**
Design a network capable of handling increased traffic and new device integrations without compromising performance.
5. **Remote Accessibility**
Enable IT administrators to securely manage and troubleshoot the network remotely.
6. **Energy and Cost Efficiency**
Consolidate multiple server rooms into a unified, virtualized system, reducing electricity consumption and maintenance expenses.
7. **Future-Readiness**
Build a flexible network structure that can integrate emerging technologies like IoT, AI-driven analytics, and smart campus sensors.

2. Literature Review

Research in recent years has shown significant advancement in campus networking technologies. The following literature provides theoretical and practical insights that directly influence the CNS project design.

Johnson & Ramirez (2022) discussed how educational institutions are adopting smart campus architectures combining LAN, WLAN, and cloud-based systems. They emphasized the role of **network resilience, centralized control, and unified communication systems**—all integral to the CNS approach.

Bennett & Hall (2022) conducted a large-scale deployment of Wi-Fi 6E across the University of Michigan campus. Their study found that Wi-Fi 6E reduced latency and improved user density management, enabling thousands of concurrent connections. This reinforces the importance of using next-generation wireless technologies in educational setups.

Liu, Chen, & Zhao (2023) performed an in-depth evaluation of Wi-Fi 6, highlighting its advantages in throughput and energy efficiency. Their work supports the integration of advanced wireless standards into the CNS network for mobile devices and smart classrooms.

Álvarez, Torres, & Fernández (2023) explored the application of Software-Defined Networking (SDN) in VLAN environments. They demonstrated that SDN could dynamically allocate resources, balance traffic loads, and enhance security through programmable network controls. While the CNS project currently relies on VLANs and RIP, it could easily evolve toward SDN-based automation in the future.

Khan & Robertson (2023) reviewed cloud-managed WLANs that use centralized controllers for multiple buildings. This aligns with the CNS model's adoption of Cisco UCS for managing both physical and virtual devices across campus.

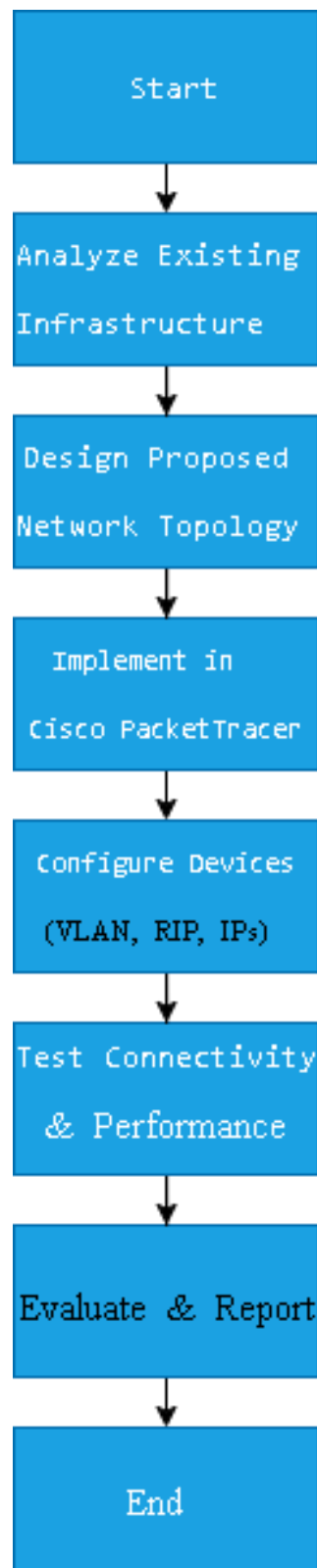
Rodriguez & Patel (2023) focused on the deployment of Microsoft SQL Server on Cisco UCS servers. Their findings show improved virtualization, simplified administration, and reduced power usage—all of which justify the CNS design's UCS-based server consolidation.

Williams & Carter (2025) analyzed a real-world university migration to Cisco UCS X-Series. Their case study reported measurable improvements in performance and cost savings, validating the feasibility of CNS's proposed infrastructure.

Together, these studies present a comprehensive understanding of the transition from traditional LAN systems to modern, cloud-ready, and software-driven campus networks.

3. Methodology

The overall methodology is explained in figure.



3.2 Explanation – Analyze Existing Infrastructure

The current college network was studied to identify limitations such as a flat network structure, outdated switches, lack of VLANs, no dynamic IP allocation, absence of firewall protection, and inefficient use of multiple server rooms. These drawbacks highlighted the need for a redesigned, centralized, and secure network.

3.3 Explanation – Design Proposed Network Topology

After analyzing the limitations, a new topology is designed using Cisco Packet Tracer. The proposed structure connects all departments through routers and switches using VLAN segmentation to enhance both security and performance. The topology includes:

- **Layer-3 Routing:** Implementation of RIP enables automatic path discovery between departmental subnets.
- **IP Subnetting:** Each department receives its own subnet, simplifying traffic isolation and troubleshooting.
- **Centralized Control:** Cisco UCS provides unified monitoring of servers and switches.
- **Server Integration:** Web, DNS, and FTP servers are configured to provide essential academic services.
- **Wireless Expansion:** Cisco Aironet access points are incorporated to offer seamless Wi-Fi connectivity throughout the campus.

3.4 Requirement Gathering

Network requirements were defined based on institutional needs, including:

- Minimum internet bandwidth of 5 Mbps for upload and download
- Scalability for future expansion
- Support for remote access
- Secure data centers and centralized server management

3.5 Network Design Planning

A hierarchical LAN architecture was planned using:

- VLANs to logically separate departments (IT, Computer, Office, Exam Cell, Internet Lab, Principal Room)
- Layer 3 IP-based switches to enable inter-VLAN routing
- Centralized server room hosting DNS, FTP, and Web servers
- Cisco-based infrastructure for reliability and enterprise-level features

3.6 IP Addressing and Routing

- A structured IP addressing plan was created using private IP ranges for different departments.
- Routing Information Protocol (RIP) was implemented to enable dynamic routing between networks.

• **IP Addressing Plan**

IT DEPARTMENT (192.168.1.0)	
HOD CABIN	192.168.1.2
IT LAB 1	192.168.1.3
IT LAB 2	192.168.1.4
IT LAB 3	192.168.1.5
IT LAB 4	192.168.1.6
Printer 0	192.168.1.7

COMPUTER DEPARTMENT (192.168.2.0)	
CS HOD CABIN	192.168.2.2
CS LAB 1	192.168.2.3
CS LAB 2	192.168.2.4
CS LAB 3	192.168.2.5
CS LAB 4	192.168.2.6
Printer 7	192.168.2.7

OTHERS (192.168.3.0)	
OFFICE	192.168.3.2
Printer 2	192.168.3.6
EXAM CELL	192.168.3.3
Printer 3	192.168.3.7
ENQUIRY	192.168.3.4
TPO	192.168.3.5
Printer 4	192.168.3.8

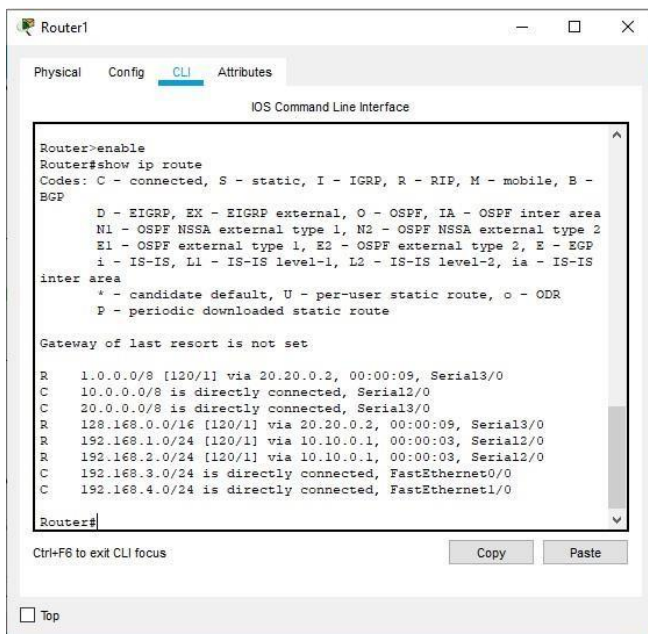
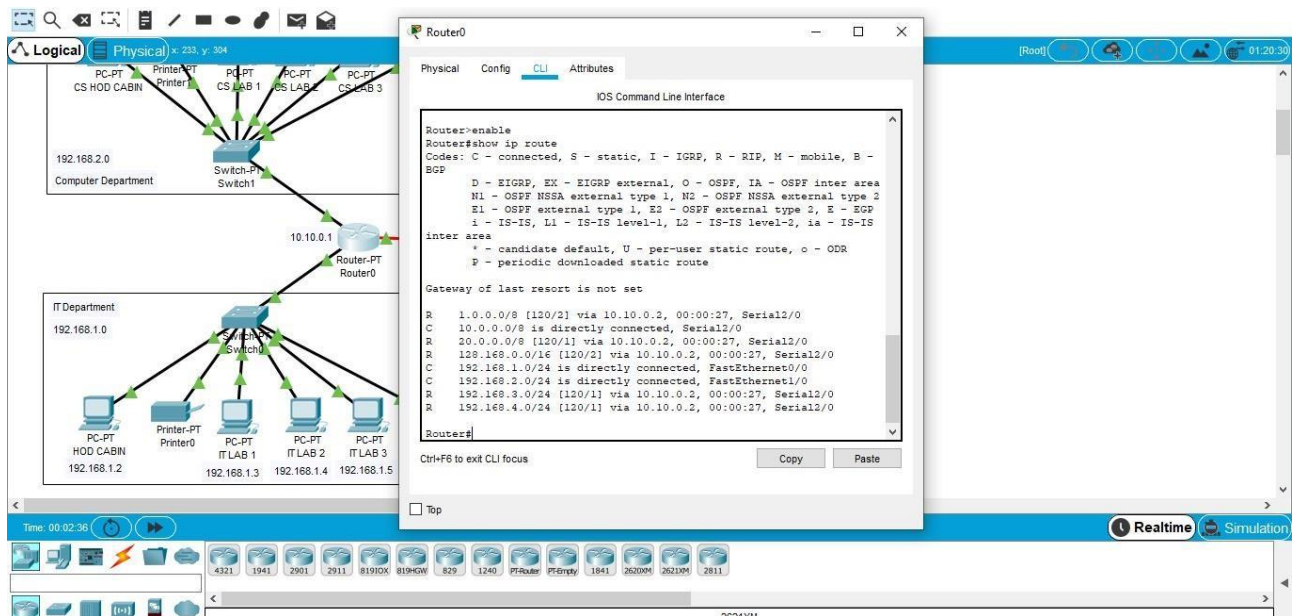
SERVER ROOM (1.0.0.0)	
FTP SERVER	1.0.0.4
PC1	1.0.0.5
DNS SERVER	1.0.0.2
WEB SERVER	1.0.0.3

INTERNET LAB (128.168.0.0)	
PC2	128.168.0.2
PC3	128.168.0.3
PC4	128.168.0.4
PC5	128.168.0.5
Printer 5	128.168.0.6

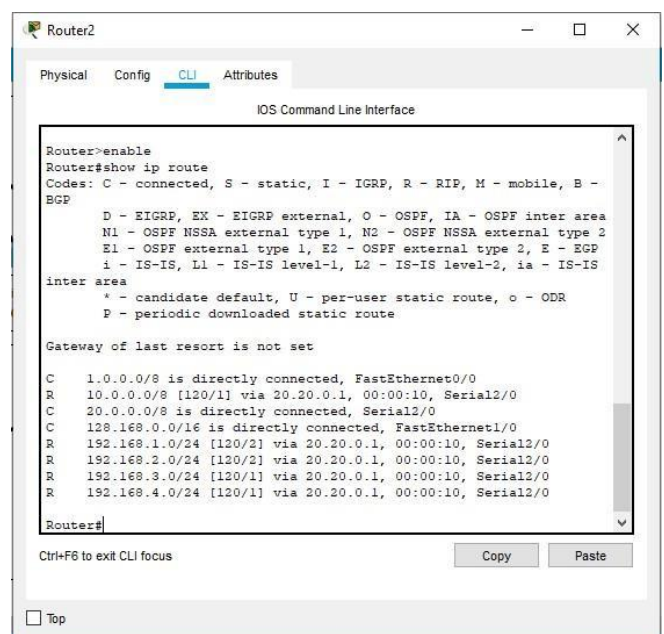
PRINCIPLE ROOM (192.168.4.0)	
PC 0	192.168.4.2
LAPTOP 0	192.168.4.3

3.7 Routing Protocol Plan

Routing Information Protocol (RIP) is a dynamic routing protocol which uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance vector routing protocol which has AD value 120 and works on the application layer of OSI model.



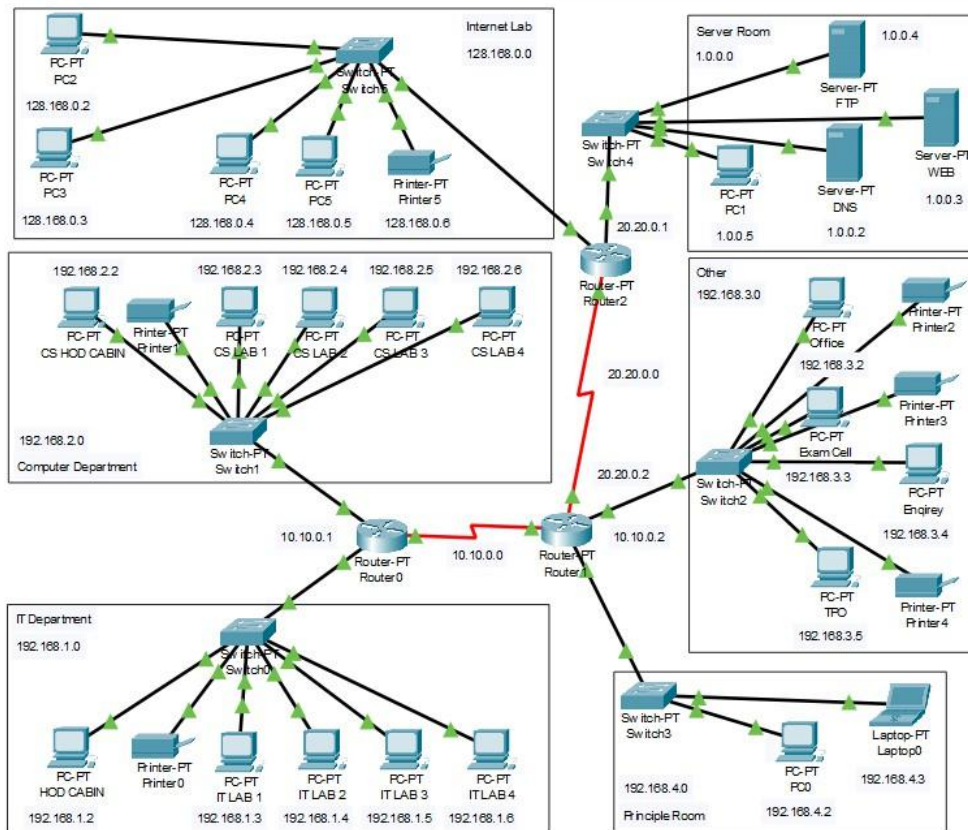
Routing Protocol Plan for Router1



Routing Protocol Plan for Router2

3.8 Simulation and Implementation

The complete network topology was implemented and simulated using **Cisco Packet Tracer**. VLAN configurations, routing protocols, and server services were configured and validated.



3.9 Testing and Validation

Network performance was tested by:

- Verifying inter-VLAN communication
- Testing access from departmental nodes to servers
- Ensuring connectivity between different network segments

HOD CABIN

Physical Config Desktop Programming Attributes

Command Prompt

```
Request timed out.
Reply from 128.168.0.4: bytes=32 time=36ms TTL=125
Reply from 128.168.0.4: bytes=32 time=38ms TTL=125
Reply from 128.168.0.4: bytes=32 time=11ms TTL=125

Ping statistics for 128.168.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 38ms, Average = 28ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=13ms TTL=126
Reply from 192.168.3.3: bytes=32 time=10ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 13ms, Average = 8ms

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.2: bytes=32 time=27ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 27ms, Average = 10ms

C:\>z
```

☐ Top

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 192.168.2.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.4: bytes=32 time=1ms TTL=126
Reply from 192.168.2.4: bytes=32 time=26ms TTL=126
Reply from 192.168.2.4: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 26ms, Average = 16ms

C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Request timed out.
Reply from 1.0.0.4: bytes=32 time=11ms TTL=126
Reply from 1.0.0.4: bytes=32 time=13ms TTL=126
Reply from 1.0.0.4: bytes=32 time=1ms TTL=126

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 8ms

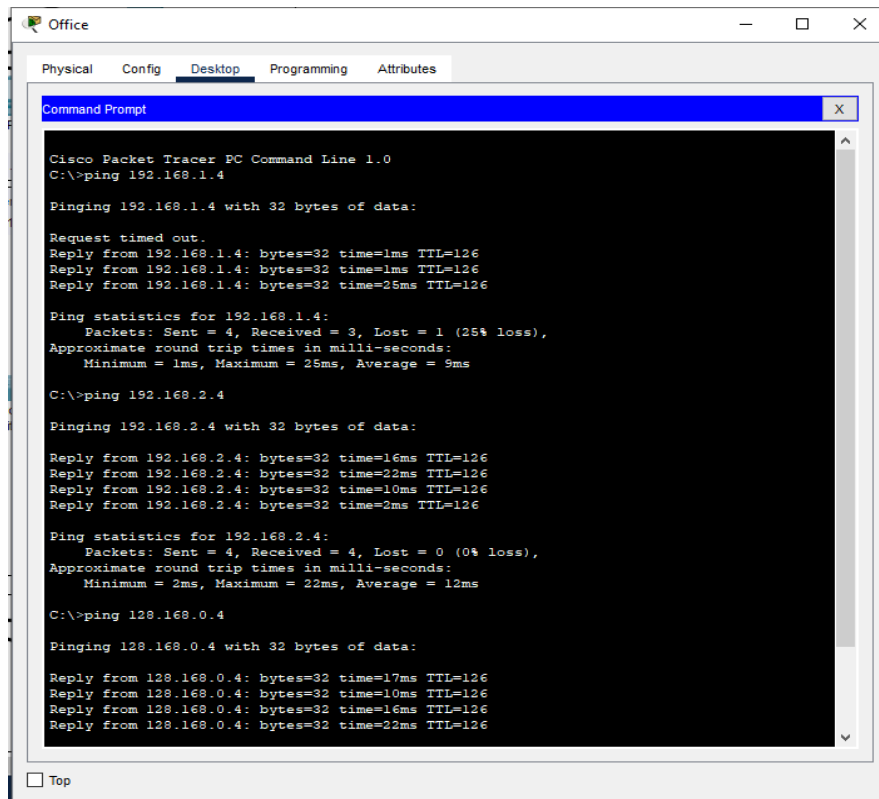
C:\>ping 128.168.0.6

Pinging 128.168.0.6 with 32 bytes of data:

Request timed out.
Reply from 128.168.0.6: bytes=32 time=1ms TTL=126
Reply from 128.168.0.6: bytes=32 time=21ms TTL=126
Reply from 128.168.0.6: bytes=32 time=16ms TTL=126

Ping statistics for 128.168.0.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
```

☐ Top



```

Office
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.4: bytes=32 time=1ms TTL=126
Reply from 192.168.1.4: bytes=32 time=1ms TTL=126
Reply from 192.168.1.4: bytes=32 time=25ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 9ms

C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=16ms TTL=126
Reply from 192.168.2.4: bytes=32 time=22ms TTL=126
Reply from 192.168.2.4: bytes=32 time=10ms TTL=126
Reply from 192.168.2.4: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 12ms

C:\>ping 128.168.0.4

Pinging 128.168.0.4 with 32 bytes of data:

Reply from 128.168.0.4: bytes=32 time=17ms TTL=126
Reply from 128.168.0.4: bytes=32 time=10ms TTL=126
Reply from 128.168.0.4: bytes=32 time=16ms TTL=126
Reply from 128.168.0.4: bytes=32 time=22ms TTL=126
  
```

4. Results:

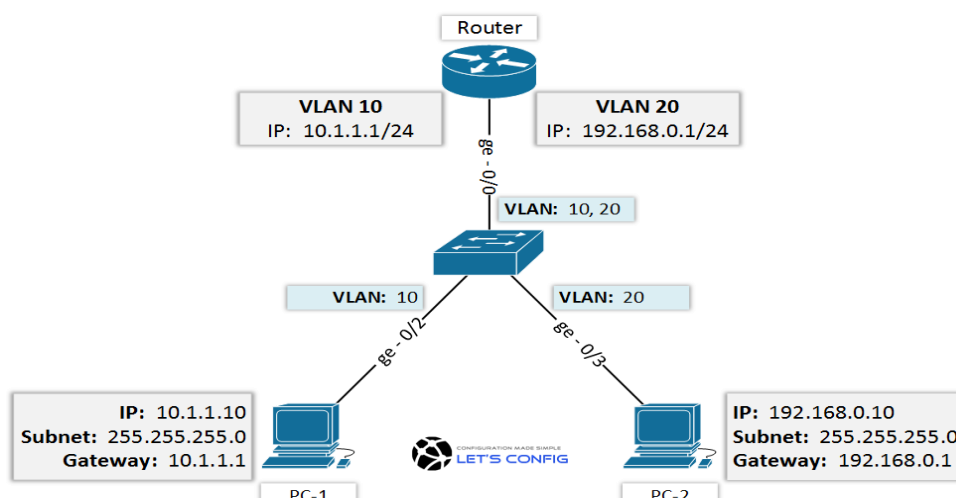
The proposed College Network Scenario was successfully designed and tested using Cisco Packet Tracer, yielding the following results:

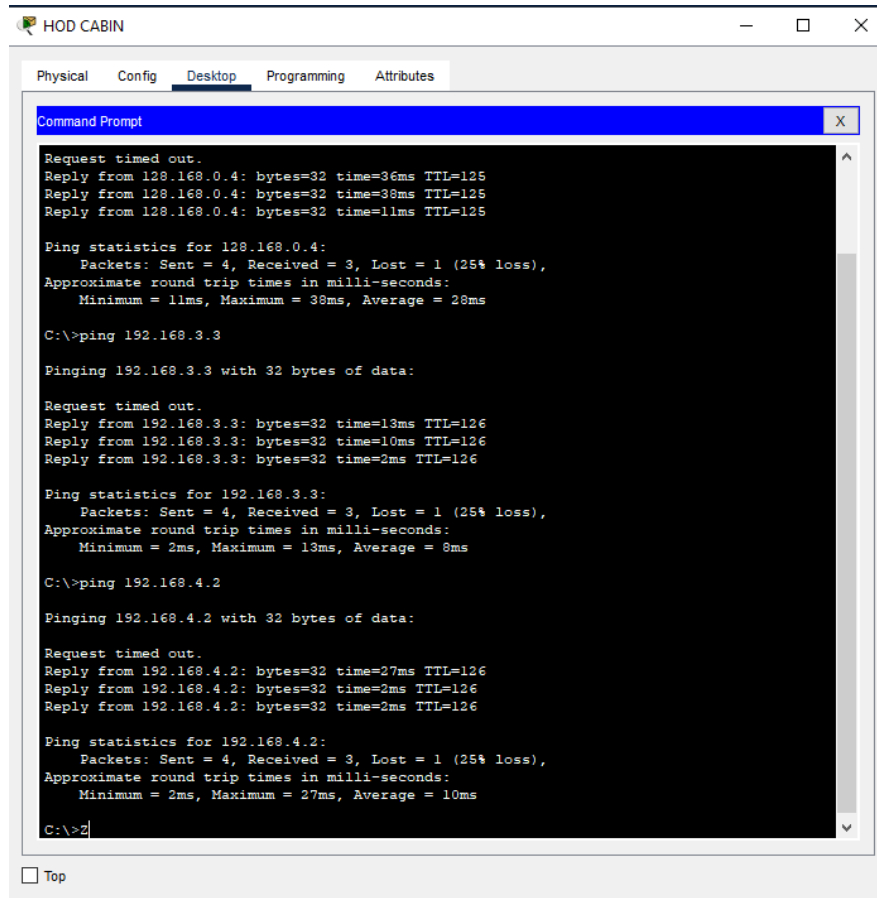
1. **Improved Network Segmentation**

VLAN implementation effectively separated departments, reducing broadcast traffic and improving network security and manageability.

2. **Successful Inter-VLAN Communication**

Devices from different departments were able to communicate efficiently through Layer 3 routing using RIP, confirming correct routing configuration.





```

HOD CABIN
Physical Config Desktop Programming Attributes

Command Prompt

Request timed out.
Reply from 128.168.0.4: bytes=32 time=36ms TTL=125
Reply from 128.168.0.4: bytes=32 time=38ms TTL=125
Reply from 128.168.0.4: bytes=32 time=11ms TTL=125

Ping statistics for 128.168.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 38ms, Average = 28ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=13ms TTL=126
Reply from 192.168.3.3: bytes=32 time=10ms TTL=126
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 13ms, Average = 8ms

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

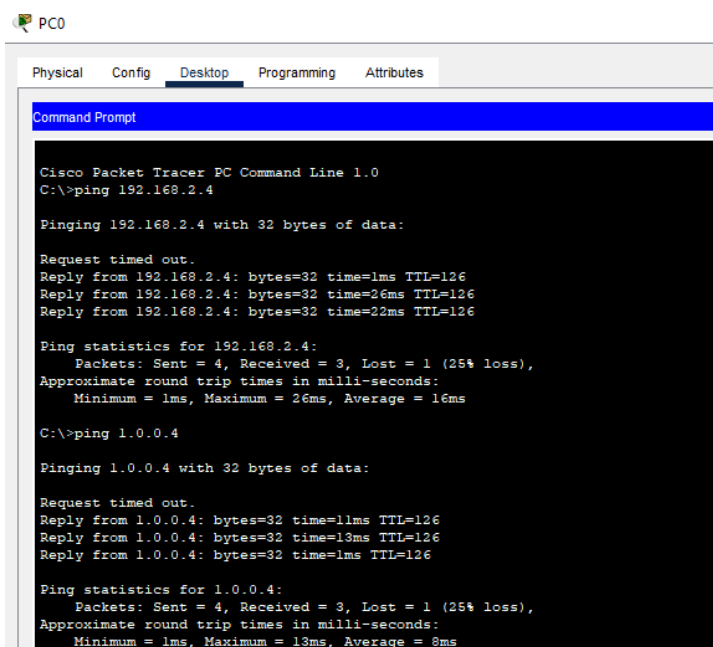
Request timed out.
Reply from 192.168.4.2: bytes=32 time=27ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 27ms, Average = 10ms

C:\>2
  
```

3. Centralized and Secure Server Access

The DNS, Web, and FTP servers hosted in a centralized server room were accessible from all authorized departments, ensuring reliable service delivery.



```

PC0
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.4: bytes=32 time=1ms TTL=126
Reply from 192.168.2.4: bytes=32 time=26ms TTL=126
Reply from 192.168.2.4: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 26ms, Average = 16ms

C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Request timed out.
Reply from 1.0.0.4: bytes=32 time=11ms TTL=126
Reply from 1.0.0.4: bytes=32 time=13ms TTL=126
Reply from 1.0.0.4: bytes=32 time=1ms TTL=126

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 8ms
  
```

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Request timed out.
Reply from 1.0.0.3: bytes=32 time=1ms TTL=126
Reply from 1.0.0.3: bytes=32 time=1ms TTL=126
Reply from 1.0.0.3: bytes=32 time=1ms TTL=126

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>1.0.0.2
Invalid Command.

C:\>ping 1.0.0.2

Pinging 1.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 1.0.0.2: bytes=32 time=1ms TTL=126
Reply from 1.0.0.2: bytes=32 time=12ms TTL=126
Reply from 1.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 1.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 4ms

C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Request timed out.
Reply from 1.0.0.4: bytes=32 time=1ms TTL=126
Reply from 1.0.0.4: bytes=32 time=16ms TTL=126
Reply from 1.0.0.4: bytes=32 time=23ms TTL=126

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 23ms, Average = 20ms

C:\>
```

IT LAB 1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Reply from 1.0.0.4: bytes=32 time=2ms TTL=125
Reply from 1.0.0.4: bytes=32 time=10ms TTL=125
Reply from 1.0.0.4: bytes=32 time=34ms TTL=125
Reply from 1.0.0.4: bytes=32 time=24ms TTL=125

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 34ms, Average = 17ms

C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Reply from 1.0.0.3: bytes=32 time=17ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125
Reply from 1.0.0.3: bytes=32 time=42ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 42ms, Average = 20ms

C:\>ping 1.0.0.2

Pinging 1.0.0.2 with 32 bytes of data:

Reply from 1.0.0.2: bytes=32 time=26ms TTL=125
Reply from 1.0.0.2: bytes=32 time=11ms TTL=125
Reply from 1.0.0.2: bytes=32 time=39ms TTL=125
Reply from 1.0.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 1.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 39ms, Average = 19ms

C:\>D
```

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Reply from 1.0.0.4: bytes=32 time=2ms TTL=125
Reply from 1.0.0.4: bytes=32 time=12ms TTL=125
Reply from 1.0.0.4: bytes=32 time=11ms TTL=125
Reply from 1.0.0.4: bytes=32 time=11ms TTL=125

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 9ms

C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Reply from 1.0.0.3: bytes=32 time=29ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125
Reply from 1.0.0.3: bytes=32 time=40ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 40ms, Average = 22ms

C:\>ping 1.0.0.2

Pinging 1.0.0.2 with 32 bytes of data:

Reply from 1.0.0.2: bytes=32 time=50ms TTL=125
Reply from 1.0.0.2: bytes=32 time=11ms TTL=125
Reply from 1.0.0.2: bytes=32 time=31ms TTL=125
Reply from 1.0.0.2: bytes=32 time=22ms TTL=125

Ping statistics for 1.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 50ms, Average = 28ms

C:\>A|
```

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Reply from 1.0.0.4: bytes=32 time<1ms TTL=127
Reply from 1.0.0.4: bytes=32 time<1ms TTL=127
Reply from 1.0.0.4: bytes=32 time<1ms TTL=127
Reply from 1.0.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 1.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Reply from 1.0.0.3: bytes=32 time<1ms TTL=127
Reply from 1.0.0.3: bytes=32 time<1ms TTL=127
Reply from 1.0.0.3: bytes=32 time<1ms TTL=127
Reply from 1.0.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 1.0.0.2

Pinging 1.0.0.2 with 32 bytes of data:

Reply from 1.0.0.2: bytes=32 time<1ms TTL=127
Reply from 1.0.0.2: bytes=32 time<1ms TTL=127
Reply from 1.0.0.2: bytes=32 time<1ms TTL=127
Reply from 1.0.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 1.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

4. Efficient Network Performance

The redesigned topology reduced congestion and improved data transfer efficiency compared to the existing flat network.

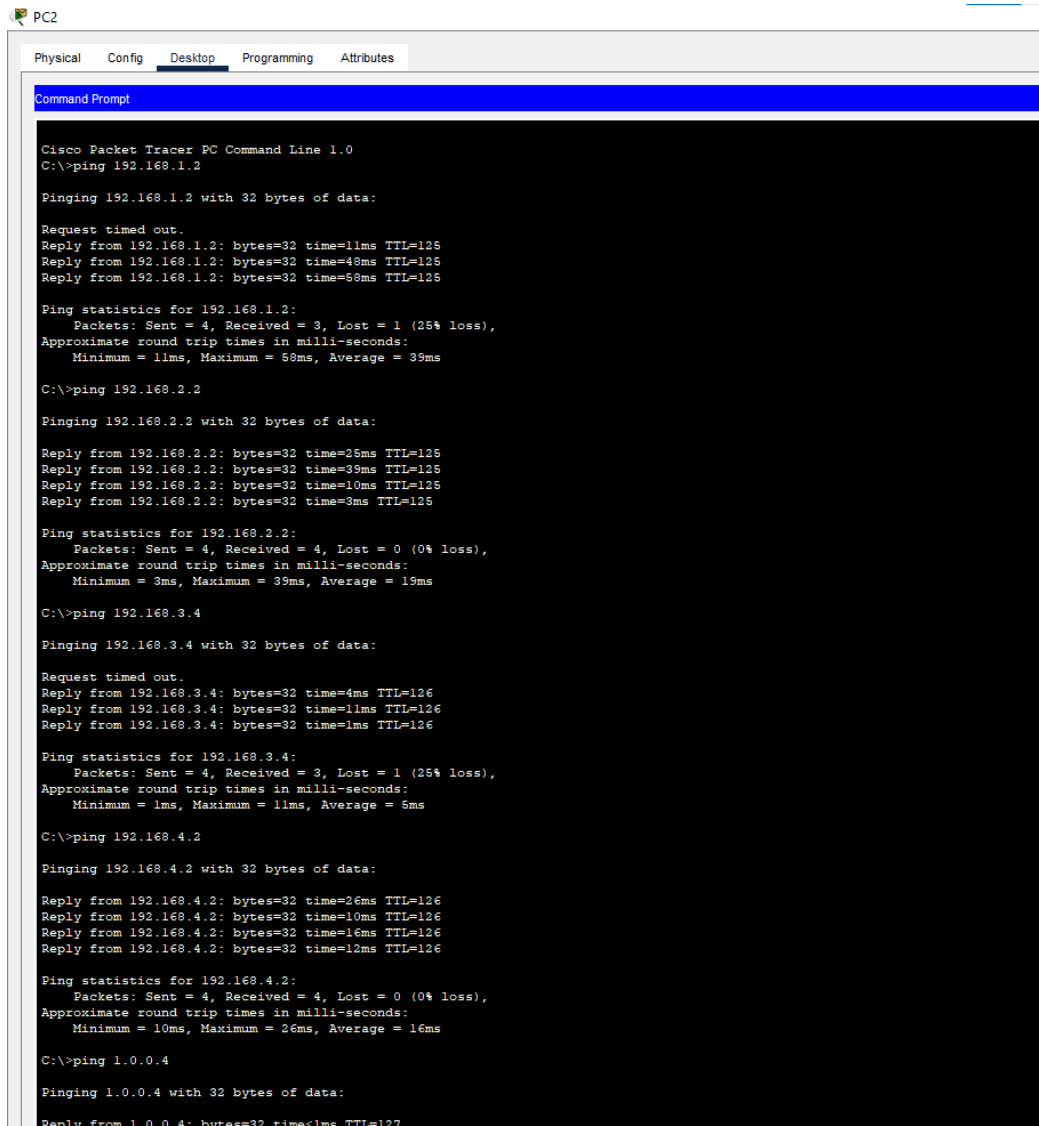
5. Scalability and Flexibility

The network design supports easy expansion, allowing additional departments, devices, and services to be integrated with minimal configuration changes.

6. Enhanced Security and Management

Use of enterprise-grade Cisco devices, VLANs, and centralized infrastructure improved monitoring, control, and overall network security.

7. Successful Service Testing



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=11ms TTL=125
Reply from 192.168.1.2: bytes=32 time=48ms TTL=125
Reply from 192.168.1.2: bytes=32 time=58ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 58ms, Average = 39ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=25ms TTL=125
Reply from 192.168.2.2: bytes=32 time=39ms TTL=125
Reply from 192.168.2.2: bytes=32 time=10ms TTL=125
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 39ms, Average = 19ms

C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.4: bytes=32 time=4ms TTL=126
Reply from 192.168.3.4: bytes=32 time=11ms TTL=126
Reply from 192.168.3.4: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 5ms

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time=26ms TTL=126
Reply from 192.168.4.2: bytes=32 time=10ms TTL=126
Reply from 192.168.4.2: bytes=32 time=16ms TTL=126
Reply from 192.168.4.2: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 26ms, Average = 16ms

C:\>ping 1.0.0.4

Pinging 1.0.0.4 with 32 bytes of data:

Reply from 1.0.0.4: bytes=32 time<1ms TTL=127
```

CS LAB 1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 128.168.0.3

Pinging 128.168.0.3 with 32 bytes of data:

Reply from 128.168.0.3: bytes=32 time=2ms TTL=125
Reply from 128.168.0.3: bytes=32 time=11ms TTL=125
Reply from 128.168.0.3: bytes=32 time=10ms TTL=125
Reply from 128.168.0.3: bytes=32 time=35ms TTL=125

Ping statistics for 128.168.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 35ms, Average = 14ms

C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Reply from 1.0.0.3: bytes=32 time=25ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125
Reply from 1.0.0.3: bytes=32 time=11ms TTL=125
Reply from 1.0.0.3: bytes=32 time=41ms TTL=125

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 41ms, Average = 22ms

C:\>ping 192.168.1.5
Invalid Command.

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.5: bytes=32 time<1ms TTL=127
Reply from 192.168.1.5: bytes=32 time<1ms TTL=127
Reply from 192.168.1.5: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 22ms, Average = 8ms

C:\>
```

IT LAB 3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 128.168.0.3

Pinging 128.168.0.3 with 32 bytes of data:

Reply from 128.168.0.3: bytes=32 time=27ms TTL=125
Reply from 128.168.0.3: bytes=32 time=41ms TTL=125
Reply from 128.168.0.3: bytes=32 time=13ms TTL=125
Reply from 128.168.0.3: bytes=32 time=10ms TTL=125

Ping statistics for 128.168.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 41ms, Average = 22ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=10ms TTL=127
Reply from 192.168.2.3: bytes=32 time=10ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 6ms

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 1.0.0.2

Pinging 1.0.0.2 with 32 bytes of data:

Reply from 1.0.0.2: bytes=32 time=29ms TTL=125
Reply from 1.0.0.2: bytes=32 time=28ms TTL=125
Reply from 1.0.0.2: bytes=32 time=18ms TTL=125
Reply from 1.0.0.2: bytes=32 time=11ms TTL=125

Ping statistics for 1.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 29ms, Average = 21ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=20ms TTL=126
```

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Ping statistics for 128.168.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 23ms, Average = 20ms

C:\>ping 128.168.0.5

Pinging 128.168.0.5 with 32 bytes of data:

Request timed out.
Reply from 128.168.0.5: bytes=32 time=15ms TTL=126
Reply from 128.168.0.5: bytes=32 time=22ms TTL=126
Reply from 128.168.0.5: bytes=32 time=34ms TTL=126

Ping statistics for 128.168.0.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 34ms, Average = 23ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=18ms TTL=126
Reply from 192.168.2.2: bytes=32 time=2ms TTL=126
Reply from 192.168.2.2: bytes=32 time=10ms TTL=126
Reply from 192.168.2.2: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 15ms, Average = 12ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=15ms TTL=126
Reply from 192.168.1.2: bytes=32 time=16ms TTL=126
Reply from 192.168.1.2: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 21ms, Average = 15ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Office
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 128.168.0.6

Pinging 128.168.0.6 with 32 bytes of data:

Request timed out.
Reply from 128.168.0.6: bytes=32 time=26ms TTL=126
Reply from 128.168.0.6: bytes=32 time=25ms TTL=126
Reply from 128.168.0.6: bytes=32 time=18ms TTL=126

Ping statistics for 128.168.0.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 26ms, Average = 23ms

C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.4: bytes=32 time=25ms TTL=126
Reply from 192.168.2.4: bytes=32 time=17ms TTL=126
Reply from 192.168.2.4: bytes=32 time=21ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 25ms, Average = 21ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=20ms TTL=126
Reply from 192.168.1.5: bytes=32 time=1ms TTL=126
Reply from 192.168.1.5: bytes=32 time=1ms TTL=126
Reply from 192.168.1.5: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.1.5:
```

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Contribution:

Member	Registration No.	Assigned Sections
Syeda Nalain Zaineb Bukhari	24108233	Introduction, Background, and Objectives, Literature Review and Research Work
Tehniat Khan	24108235	Network Design (Topology, VLANs, RIP concepts)
Aiman Nasir	24108202	Implementation in Cisco Packet Tracer and Configuration
Malika Naz	24108212	Testing, Evaluation, and Expected Outcomes
Safia Alam	24108229	Documentation, Formatting, and Final Report Compilation